

FORT-DE-FRANCE, Martinique

WMO: 789250

Lat: 14.595N

Lon: 60.996W

Elev: 26

StdP: 14.68

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00404

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	67.9	69.7	62.9	85.8	76.0	64.3	90.4	76.6	23.3	81.9	22.1	82.0	3.0	100	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	90.0	78.4	89.6	78.3	88.3	78.0	80.4	87.1	80.0	86.5	79.6	86.0	11.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	149.9	84.3	78.0	145.8	83.6	77.4	142.4	83.1	44.1	87.3	43.5	87.1	43.0	86.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	19.4	18.1	64.2	92.5	1.9	1.2	62.8	93.4	61.8	94.0	60.7	94.7	59.4	95.5		
(5)			62.8	82.0	1.6	1.2	61.7	82.8	60.7	83.5	59.8	84.2	58.6	85.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.5	78.1	78.0	78.6	79.9	81.4	82.2	82.2	82.3	82.1	81.6	80.5	79.1	
	DBStd	2.25	1.54	1.61	1.76	1.73	1.56	1.36	1.37	1.48	1.49	1.54	1.59	1.59	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11142	872	783	887	897	975	965	999	1002	964	979	916	903	
	CDD65	5667	407	363	422	447	510	515	534	537	514	514	466	438	
	CDH74	54067	2977	2640	3342	4030	5239	5604	5843	5839	5447	5134	4307	3665	
	CDH80	16211	547	517	806	1138	1635	1730	1861	2046	2044	1791	1272	824	
Wind	WSAvg	9.0	9.6	10.4	9.6	9.3	9.7	11.1	10.5	8.2	6.7	6.7	7.3	9.0	
Precipitation	PrecAvg	87.6	5.0	3.3	3.7	4.6	6.1	7.8	9.7	11.8	9.4	9.4	9.7	6.8	
	PrecMax	120.3	7.8	7.8	7.3	16.5	14.5	13.5	14.9	30.8	16.3	18.7	21.8	14.8	
	PrecMin	69.3	2.0	0.4	1.4	1.3	2.0	3.7	4.1	6.2	4.0	3.7	2.2	3.0	
	PrecStd	14.6	1.5	1.4	1.7	3.6	3.5	2.6	2.9	4.9	3.2	4.1	4.6	2.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	87.3	88.2	89.7	89.9	89.7	89.5	90.8	91.7	91.4	89.6	87.5	
		MCWB	74.7	75.0	75.3	76.9	78.0	78.5	78.7	79.3	78.9	78.7	78.4	76.6	
	2%	DB	84.5	85.1	86.3	87.9	88.2	88.1	88.1	89.6	90.0	89.7	88.0	86.1	
		MCWB	74.2	73.8	74.5	76.2	77.1	77.8	78.2	78.7	78.5	78.4	78.0	76.2	
	5%	DB	84.0	84.2	85.2	86.3	87.7	87.6	87.8	88.2	89.5	88.2	87.1	84.7	
		MCWB	74.1	73.4	74.0	75.3	76.9	77.6	78.0	78.5	78.4	78.3	77.7	75.4	
	10%	DB	82.8	82.8	84.2	85.7	86.4	86.4	86.6	87.8	88.1	87.7	86.1	84.2	
		MCWB	73.6	72.9	73.5	75.0	76.4	77.2	77.7	78.4	78.4	78.2	77.4	75.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.0	76.5	77.3	78.5	80.0	80.3	80.6	81.0	81.0	80.9	80.1	78.8
			MCDB	82.8	83.9	84.6	85.8	86.8	86.5	86.7	87.2	88.0	87.1	86.6	84.4
2%		WB	76.0	75.4	76.0	77.4	79.0	79.6	79.8	80.1	80.1	80.0	79.3	77.8	
		MCDB	81.9	82.4	83.4	84.6	85.7	85.6	85.9	86.6	87.2	86.6	85.7	83.6	
5%		WB	75.3	74.6	75.2	76.6	78.2	78.9	79.2	79.7	79.7	79.5	78.7	77.0	
		MCDB	81.3	81.4	82.6	83.7	84.8	84.8	85.2	86.0	86.7	86.1	84.9	82.8	
10%		WB	74.6	74.0	74.6	75.9	77.4	78.2	78.6	79.1	79.1	78.9	78.0	76.2	
		MCDB	80.6	80.7	81.6	82.8	84.0	84.1	84.5	85.3	85.9	85.5	83.9	81.9	
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.5	12.0	12.0	11.0	9.6	10.0	11.4	13.1	12.8	12.0	11.2	
		MCDBR	11.8	12.7	13.2	13.1	12.0	10.3	10.5	12.2	14.6	13.9	13.2	11.8	
		MCWBR	5.2	5.2	5.1	4.9	4.4	4.1	4.3	4.9	5.6	5.7	5.6	5.3	
	5% WB	MCDBR	10.6	11.4	12.0	12.2	11.2	9.6	10.0	11.6	13.1	13.0	12.1	11.1	
		MCWBR	5.2	5.3	5.1	5.0	4.8	4.5	4.7	5.2	5.7	5.9	5.7	5.4	
Clear-Sky Solar Irradiance	taub	0.366	0.381	0.380	0.416	0.464	0.550	0.517	0.462	0.424	0.385	0.377	0.376		
	taud	2.607	2.539	2.549	2.469	2.323	2.038	2.144	2.358	2.490	2.625	2.630	2.601		
	Ebn at Noon	296	295	296	283	266	241	250	267	279	290	290	289		
	Edh at Noon	30	33	34	36	41	54	49	40	35	30	29	30		
All-Sky Solar Radiation	RadAvg	1628	1847	1994	2043	1987	1925	1969	1990	1892	1709	1570	1540		
	RadStd	48	41	85	88	127	96	84	96	85	87	113	58		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.54	+1.22	N/A	+0.63	N/A	N/A	N/A	+203	+202
(47) Regional (2 neighbors)		N/A	+0.86	N/A	N/A	N/A	N/A	N/A	+203	+202

Nomenclature: See separate page